

KV Cache Memory Wall memory-bandwidth-bound decode

Algorithmic compression (§4)

Quantization · Eviction / sparsity · Token & cross-layer merging

Architectural redesign (§5)

MQA / GQA · Multi-head latent attention · SSM & hybrid models

System-level management (§6)

Paged attention · Prefix sharing (+ security) · Tiered offloading

Hardware acceleration (§7)

Decode kernels & fusion · Processing-in-memory · ASIC / FPGA

Cross-cutting (§8–§9): co-design & Pareto trade-offs · infinite context · agentic degradation · benchmarking · multi-tenant security